

Praneetha

Data Scientist

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A Data Scientist with around 7 years of experience turning complex data into actionable intelligence, specializing in predictive modeling, NLP, and compliance-driven analytics across finance, healthcare, retail, and insurance. I build scalable data solutions that not only solve technical challenges but also drive measurable improvements in business performance, regulatory compliance, and customer outcomes.

PROFESSIONAL SUMMARY

- Around 7 years of experience delivering enterprise-scale data solutions, advancing from analyst to data scientist roles, with proven expertise in ETL/ELT pipelines, machine learning, NLP, and advanced analytics applied to retail, banking, healthcare, and insurance.
- Designed and deployed enterprise-scale ETL/ELT pipelines and data workflows, integrating multi-source datasets into Snowflake, Redshift, BigQuery, and Hadoop ecosystems to support real-time analytics and regulatory reporting.
- Built and deployed machine learning models for credit risk scoring, loan default prediction, claims fraud detection, customer churn prevention, and patient outcome forecasting, improving accuracy and reducing operational risks.
- Developed NLP pipelines to extract and structure data from unstructured clinical and financial text sources, including physician notes, claims reports, and CRM logs, enabling deeper analytics and compliance alignment.
- Automated reporting frameworks for Basel III, IFRS 17, Solvency II, AML/KYC, HIPAA, and HITECH, reducing manual reporting time by 30–40% and improving audit readiness.
- Created interactive BI dashboards (Power BI, Tableau, Qlik) to monitor KPIs such as NPA ratios, hospital occupancy, claims settlement times, Net Interest Margin, and customer retention, enabling faster and more informed executive decision-making.
- Experienced in cloud platforms (AWS, Azure, GCP) and big data ecosystems (Spark, Hadoop, Kafka), with proven success in deploying large-scale solutions that enhance efficiency, compliance, and forecasting accuracy.
- Strong communicator with a track record of collaborating across business, actuarial, clinical, and compliance teams, ensuring data-driven strategies directly support organizational goals.

TECHNICAL SKILLS

Programming Language	Python (Pandas, NumPy, scikit-learn, TensorFlow, Keras, PyTorch, SpaCy, NLTK), R, SQL, PL/SQL
Machine Learning & AI	Supervised & Unsupervised Learning, Deep Learning (CNN, LSTM), Time-Series Forecasting (ARIMA, SARIMA, Prophet), Clustering, Classification, Regression, Survival Analysis, Hypothesis Testing, EDA
Natural Language Processing	Text Mining, NER, Topic Modeling, Sentiment Analysis, Clinical Text Analysis, Tokenization, Lemmatization
Data Engineering & ETL	Airflow, Informatica PowerCenter/IICS, Data Cleaning & Transformation, API Integration (REST, SOAP), Snowflake, Redshift, BigQuery, PySpark, Hadoop, Hive
Databases	MySQL, MS SQL Server, PostgreSQL, Oracle, Teradata
Data Visualization	Power BI, Tableau, Qlik, Plotly, Matplotlib, Seaborn, Dashboard Development
Cloud & Deployment	AWS (S3, RDS, Redshift, SageMaker, Lambda), Azure (ADF, Synapse, ML Studio), GCP (BigQuery, Vertex AI)
Tools & IDEs	Jupyter Notebook, Anaconda, Spyder, Git, Jira
Methodologies & Compliance	Agile/Scrum, Data Governance, Metadata & MDM, HIPAA/HITECH, Basel III/AML/KYC, IFRS 17, Solvency II, GDPR, CCPA, ISO 27001

WORK EXPERIENCE

Role: Data Scientist

Feb 2025 – Present

Client: Wells Fargo, Charlotte, NC

Project Overview:

Wells Fargo, a leading multinational financial services company, initiated a **data modernization and advanced analytics program** focused on **data governance**. As a Data Scientist it involves developing **credit risk and fraud detection models**, **portfolio optimization frameworks**, and **customer retention analytics**, while consolidating diverse datasets including **transactional, CRM, claims, and market data** into unified **cloud-based data lakes**. These solutions support **regulatory reporting, compliance audits, and executive decision-making**, driving measurable improvements in forecasting accuracy, audit readiness, and overall data-driven strategy.

Technical Contributions:

- Designed and deployed **data pipelines** to ingest, clean, and transform structured and unstructured datasets from banking, insurance, and investment systems into Snowflake and Hadoop ecosystems.
- Built and validated **machine learning models** for **credit risk scoring, loan default prediction, fraud detection, AML/KYC compliance, and customer churn prevention**, improving risk detection accuracy by 15%.
- Applied **time-series forecasting models (ARIMA, LSTM, Prophet)** for predicting **interest rates, stock prices, and market volatility**; leveraged **Modern Portfolio Theory (MPT)** to optimize investment portfolios, improving returns by 8%.
- Automated compliance reporting pipelines for **Basel III, IFRS 17, Solvency II, PCI-DSS, FATF, and OFAC**, reducing manual effort by 40% and improving audit readiness.
- Incorporated **Bloomberg API, Reuters Eikon, and Moody's Analytics** feeds into market forecasting and risk models, enhancing predictive accuracy.
- Applied **actuarial and statistical analysis** for insurance pricing, claims forecasting, and fraud detection, reducing fraudulent payouts by 20%.
- Built executive-level **Power BI, Tableau, and Qlik** dashboards to monitor KPIs including **NPA ratios, Net Interest Margin (NIM), policy lapse rates, and portfolio risk exposure**, enabling faster strategic decisions.
- Ensured **GDPR, CCPA, HIPAA, and ISO 27001** compliance by implementing encryption, secure access controls, and data governance best practices.

Environment: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch), R, SQL, Spark, Hadoop, Kafka, Snowflake, AWS, Azure, SAS, Oracle, SQL Server, Power BI, Tableau, Qlik, Bloomberg API, Reuters Eikon, Moody's Analytics, Agile, CI/CD (Jenkins, GitHub), SDLC.

Role: Data Scientist

Sep 2023 – Dec 2024

Client: Celon Behavioral Health, Ashburn, VA

Project Overview:

As a Data Scientist at Celon, a leading healthcare and life sciences organization, I developed **models for patient readmission, disease progression, and treatment forecasting**, while also designing **NLP pipelines to extract entities from unstructured physician notes and pathology reports**, mapping results to **ICD-10, CPT, and LOINC codes** for seamless EHR integration. Additionally, I implemented **time-series forecasting** to predict hospital occupancy and seasonal disease trends, helping healthcare providers optimize capacity planning. All work was executed under strict adherence to **HIPAA, HITECH, and healthcare interoperability standards (HL7/FHIR)**, ensuring secure, compliant, and impactful healthcare analytics.

Technical Contributions:

- Designed, trained, and validated **predictive models** for **patient readmission risk, treatment outcomes, and disease progression** using Python (Scikit-learn, TensorFlow, Keras).
- Built **NLP pipelines** to extract entities from physician notes, pathology reports, and diagnostic summaries; implemented **NER and topic modeling** for structured EHR integration.

- Developed time-series models (ARIMA, LSTM, Prophet) for predicting hospital occupancy, seasonal disease outbreaks, and resource utilization.
- Processed and standardized EHR data (Epic, Cerner), insurance claims, and lab datasets into analytical workflows, reducing data prep time by 30%.
- Applied data balancing (SMOTE, class weighting), feature engineering, and normalization techniques to improve model accuracy on imbalanced datasets.
- Conducted EDA and statistical analysis to uncover correlations between demographics, comorbidities, and treatment outcomes.
- Designed Tableau and Power BI dashboards tracking healthcare KPIs including readmission rates, treatment effectiveness, and disease prevalence.
- Partnered with clinicians, statisticians, and IT teams to design analytical frameworks for clinical trials and research studies.
- Ensured HIPAA/HITECH compliance with encryption, de-identification, and role-based access controls.

Environment: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, spaCy, NLTK), R, SQL, Epic EHR, Cerner, HL7/FHIR standards, ICD-10, CPT, LOINC, Snowflake, SQL Server, AWS, Azure, Tableau, Power BI, Matplotlib, Seaborn, Agile SCRUM, Data Governance, HIPAA/HITECH Compliance.

Role: Data Analyst

Jul 2020 –

Jan 2023

Client: ICICI, Hyderabad India

Project Overview:

Contributed to data analytics, reporting, and BI initiatives in the banking and insurance sector, focusing on claims management, fraud detection, customer segmentation, and regulatory compliance. Designed and optimized pipelines, automated compliance reporting, and built dashboards that improved operational efficiency, supported underwriting and actuarial teams, and strengthened fraud detection processes while aligning with IRDAI regulatory requirements.

Technical Contributions:

- Designed SQL and Python (Pandas, NumPy) pipelines to collect, validate, and transform claims, transactions, and customer datasets, ensuring 98%+ accuracy in analytical models.
- Built and deployed Power BI and Tableau dashboards tracking claims settlement times, fraud detection rates, premium collections, and policy renewal ratios, enabling real-time insights for executives.
- Conducted loss ratio and claims trend analysis, identifying key retention drivers and supporting campaigns that increased customer retention by 12%.
- Automated IRDAI regulatory reports, reducing manual reporting effort by 35% and improving audit compliance.
- Supported actuarial teams by preparing analytical datasets for premium pricing and risk scoring, enhancing model accuracy and transparency.
- Developed fraud detection logic in Python to flag anomalies and high-risk claims, helping reduce fraudulent payouts and strengthen internal investigations.
- Authored optimized SQL queries, joins, and stored procedures, reducing query runtime by 20% and improving SLA adherence.
- Partnered with IT teams to streamline ETL pipelines into Snowflake, improving data refresh cycles and report latency.
- Presented analytical insights to underwriting, compliance, and fraud teams, influencing pricing strategies, portfolio monitoring, and customer engagement.
- Ensured strict compliance with IRDAI data standards, implementing secure, auditable, and high-quality data workflows.

Environment: Python (Pandas, NumPy, Matplotlib, Seaborn), R, SQL, Banking & Insurance, Risk Analytics, Claims Management, Fraud Detection, IRDAI Compliance, Snowflake, SQL Server, Stored Procedures, Power BI, Tableau, Excel, IRDAI Compliance, Claims Data, Risk Analytics, Jupyter Notebook, Git.

Role: Junior Data & Reporting Analyst

Jul 2018 –

Jun 2020

Client: Big Basket, Hyderabad India

Project Overview:

Launched career in data reporting and business intelligence within a leading e-commerce platform, focusing on supply chain optimization, customer analytics, and operational reporting. Built strong foundations in SQL, Python, and BI tools by creating automated dashboards, optimizing reporting workflows, and delivering insights that improved order fulfillment, reduced wastage, and enhanced customer retention strategies.

Technical Contributions:

- Designed and automated Power BI and Tableau dashboards to track KPIs including order fulfillment rates, delivery times, returns, and customer satisfaction scores.
- Developed SQL queries, joins, and stored procedures to extract and prepare analytical datasets, improving reporting accuracy and reducing runtime by 25%.
- Built Python (Pandas, NumPy) scripts for data transformation, feature calculation, and ad-hoc analysis, cutting manual processing time by 30%.
- Conducted EDA and statistical analysis to identify purchase patterns, seasonal demand fluctuations, and supply bottlenecks, leading to improved demand forecasting.
- Produced weekly, monthly, and quarterly reports on sales performance, customer churn, and marketing campaigns, enabling data-driven decision-making by leadership.
- Partnered with business stakeholders to gather reporting requirements and translate them into scalable BI solutions.
- Established data validation checks and version control (Git) to ensure reporting consistency, auditability, and governance.
- Delivered insights that improved order accuracy by 12% and reduced inventory wastage by 8%, strengthening supply chain efficiency.

Environment: E-Commerce, Retail Analytics, Supply Chain Optimization, Customer Insights, SQL, Python (Pandas, NumPy, Matplotlib, Seaborn), MS SQL Server, AWS S3, Power BI, Tableau, Excel, Jupyter Notebook, Git.